

The Variational Quantum Algorithm Framework

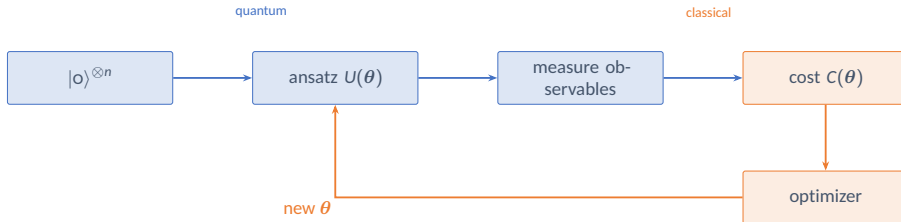
Grover adaptive search, then the variational framework — ansatz, cost, gradient, optimizer, and where it breaks

Lecture 5

September 21, 2026

Where we are

- Quantum Annealing reached the ground state by letting physics carry the state there.
- This lecture covers the two *gate-model* answers:
 - 1 First **Grover adaptive search**, which carries a provable speedup but needs a fault-tolerant machine
 - 2 **Variational** idea, which makes the circuit adjustable and lets a classical optimizer steer it



Four choices define any VQA: **the ansatz, the cost function, how gradients are obtained, and which optimizer runs the loop.** Each of the four can fail on its own.

Quantum Optimization Algorithms

- Quantum Optimization Algorithms leverage quantum physics to solve optimization problems more efficiently
- Many real-world problems are mapped into combinatorial optimization
 - Scheduling airplane flights
 - Assignments of pilot and cabine crews
 - Last-mile delivery
 - Class scheduling
 - Designing water distribution networks
 - Uber!
 - Earth and Geoscience problems

Grover Adaptive Search — the objective

The objective

Find the *exact* minimiser of $f(x) = x^T Q x$ over all 2^n bit strings, using provably fewer evaluations than checking them one at a time.

- GAS deliberately ignores the structure of f . It treats the 2^n candidates as an *unstructured list* and searches it
- Optimization becomes a sequence of **searches**, each with a tighter threshold:
 1. Pick a threshold y — start with the value of any random solution
 2. Build an oracle that marks every x with $f(x) < y$
 3. Run Grover amplitude amplification to sample such an x
 4. Set $y \leftarrow f(x)$ and repeat until no better x is found
- For a QUBO the oracle is a fixed, known circuit: evaluate f into a register in phase encoding, then test its sign
- Dürr-Høyer proved the whole loop finds the minimum of N values in $O(\sqrt{N})$

C. Dürr & P. Høyer (1996). "A Quantum Algorithm for Finding the Minimum." arXiv:quant-ph/9607014 (*preprint*) · A. Gilliam, S. Woerner, C. Goniculea (2021). "Grover Adaptive Search for Constrained Polynomial Binary Optimization." *Quantum* 5, 428. arXiv:1912.04088

Grover Adaptive Search — what it costs

- With $N = 2^n$ candidates, GAS needs $O(\sqrt{N}) = O(2^{n/2})$ oracle calls
 - a genuine, *provable* quadratic speedup — the only one in this course
 - and still exponential: $n = 100$ means 2^{50} oracle calls
- Grover's $O(\sqrt{N})$ is also *optimal* for unstructured search — no quantum algorithm does better without exploiting structure. So this is the ceiling, not a first attempt
- **The hardware bill is the real obstacle**
 - the oracle must evaluate f coherently, and amplitude amplification iterates it $O(\sqrt{N})$ times
 - the whole run must stay coherent — one long circuit, not many short ones
 - that is a fault-tolerant requirement, not a NISQ one

The catch: GAS has the proof but not the hardware. Everything else in this course has the hardware but not the proof.

Variational Quantum Algorithms (VQA)

- VQA is a class of hybrid quantum-classical algorithms that can be used to solve optimization problems
- The hybrid structure of VQA enables effective utilization of both classical and quantum resources -> well-suited for NISQ-devices
- VQA partitions the computation workload between a classical optimizer and a QPU
- VQA use parametrized quantum circuits with depths falling within the coherence time of existing NISQ devices
- VQA use iteration and parameter fine-tuning to confer robustness against quantum errors
- We will cover two VQA:
 - Variational Quantum Eigensolver
 - Quantum Approximation Optimization Algorithm

M. Cerezo et al. (2021). "Variational quantum algorithms." *Nature Reviews Physics* 3, 625–644. arXiv:2012.09265 · J. R. McClean, J. Romero, R. Babbush, A. Aspuru-Guzik (2016). *New J. Phys.* 18, 023023.

Two objectives, side by side

	Grover adaptive search	Variational (VQE, QAOA)
Objective	the <i>exact</i> minimiser	a <i>good</i> solution, quickly
Guarantee	provable $O(\sqrt{N})$	none
Uses structure of f ?	no — unstructured search	yes — through the ansatz
Circuit	one long coherent run	many short runs
Depth set by	$\sqrt{N}$ amplification steps	the ansatz, fixed by you
Classical role	none	the optimizer — half the algorithm
Output	the optimum, with high probability	a distribution; keep the best sample
Hardware era	fault-tolerant	NISQ, today
Fails when	coherence runs out	ansatz misses, or gradients vanish

- **They trade the same two things in opposite directions.** GAS keeps the guarantee and pays for it in coherence. The variational methods keep the shallow circuit and pay for it with the guarantee
- Notice what the middle rows imply: GAS cannot be helped by knowing your problem well, and a VQA cannot be helped by waiting for a better proof. Improving each one means improving a different thing

Takeaway: The rest of this lecture develops the right-hand column, because it is the one that runs on hardware you can actually book time on.

One framework, four algorithms

Everything in the second half of this course is the same loop with a different cost function bolted on. The table is worth memorising: it is the map of the rest of the term.

Instance	Ansatz	Cost $C(\theta)$	What you keep
VQE (L6)	UCC, or hardware-efficient	$\langle \psi(\theta) H \psi(\theta) \rangle$	a number: the energy
QAOA (L7)	alternating cost/mixer layers	$\langle H_C \rangle$, or a tail statistic	a bitstring: the best sample
VQLS (L6)	hardware-efficient	overlap of $A \psi \rangle$ with $ b \rangle$	a state, sampled functionals
PDE solvers (L10–L11)	as VQLS, plus a time loop	discretized residual norm	the solution field, sampled
Machine learning	encoding + variational layers	empirical loss on training data	the trained θ^*

What is genuinely shared: the parametrized circuit, the estimator that turns shots into $C(\theta)$, the gradient machinery, and every failure mode we cover today. Fix a plateau in one and you have fixed it in all five.

What is not: three of the five return a number or a sample rather than a state. That is not an implementation shortcoming — it is what measurement allows, and it decides which problems the framework can even be pointed at.

What is an ansatz?

Ansatz (German: “approach”, an educated guess): a parametrized circuit whose *structure* is fixed before optimization starts. Only its parameters are tuned.

$$|\psi(\theta)\rangle = U(\theta) |\psi_0\rangle, \quad \theta \in \mathbb{R}^m$$

Three design choices

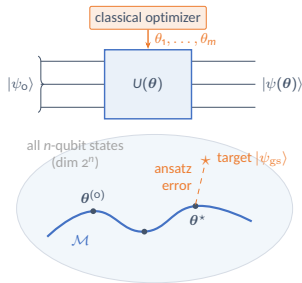
- **Reference state** $|\psi_0\rangle$: $|0\rangle^{\otimes n}$, $|+\rangle^{\otimes n}$, or Hartree-Fock
- **Gate layout**: which rotations, which entanglers, in what connectivity pattern
- **Parameters**: how many (m) and where. Usually one block repeated L times

What it decides. The ansatz fixes the *reachable set* $\mathcal{M} = \{|\psi(\theta)\rangle : \theta \in \mathbb{R}^m\}$, a family of at most m dimensions inside a 2^n -dimensional space. The optimizer only ever moves inside $\mathcal{M}$, so

$$\min_{\theta} C(\theta) \geq \min_{|\psi\rangle} \langle \psi | H | \psi \rangle,$$

with equality only when the target state lies in $\mathcal{M}$.

Takeaway: The ansatz is the model. It decides what the optimizer can find before a single shot is taken.



The optimizer walks along $\mathcal{M}$ to its best point θ^* . If the target is off the curve, no amount of optimization closes the gap.

Choosing an ansatz

Hardware-efficient

Layers of single-qubit rotations and the device's native entangler.

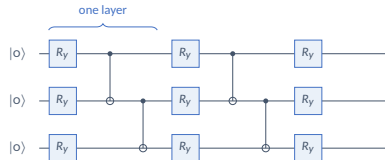
- Shallow, runs on real hardware
- Highly expressive
- Expressiveness is the problem — see *Barren plateaus*

Problem-inspired

Built from the structure of the problem: QAOA's alternation, UCC for chemistry.

- Fewer parameters, better-conditioned landscape
- Deeper, and needs domain knowledge

Takeaway: Depth is a two-sided constraint. Too shallow cannot represent the answer; too deep cannot be trained. The useful window is narrow and problem-dependent.



A hardware-efficient ansatz: rotations, entanglers, repeat.

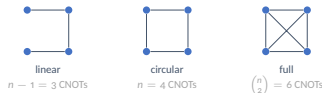
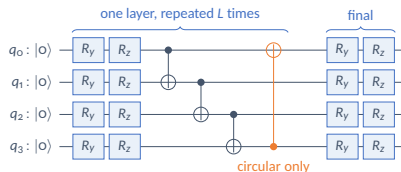
Common ansatz circuits (1): hardware-efficient

Rotation layers on every qubit, then an entangling layer of the device's native two-qubit gate. Repeat L times and finish with one more rotation layer.

Variant	Rotations	Parameters m
RealAmplitudes	R_y	$n(L + 1)$
EfficientSU2	R_y, R_z	$2n(L + 1)$

(names from the Qiskit circuit library)

- R_y only keeps every amplitude real, which is enough for real Hamiltonians such as Ising and QUBO costs
- Adding R_z reaches complex amplitudes, at twice the parameters
- **Entanglement pattern** sets the CNOTs per layer. It should match the chip's coupling map, or the compiler inserts SWAPs
- Problem-agnostic: symmetries such as particle number are not preserved



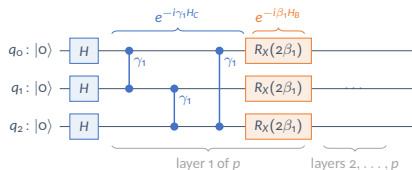
EfficientSU2 on 4 qubits. The dark CNOTs are the linear pattern; adding the orange one makes it circular.

Common ansatz circuits (2): QAOA

A *problem-inspired* ansatz, derived from the cost Hamiltonian H_C of the QUBO/Ising model (L3-L4) and the mixer $H_B = \sum_i X_i$.

$$|\gamma, \beta\rangle = \prod_{k=1}^p e^{-i\beta_k H_B} e^{-i\gamma_k H_C} |+\rangle^{\otimes n}$$

- **Reference:** $|+\rangle^{\otimes n}$, all 2^n bitstrings equally weighted
- **Cost layer:** one ZZ rotation per edge of the problem graph (CNOT, R_Z , CNOT; details in L7) and one R_Z per linear term
- **Mixer layer:** $R_X(2\beta) = e^{-i\beta X}$ on every qubit
- **Parameters:** $m = 2p$, independent of n



Max-Cut on a triangle: three edges
 $\Rightarrow$ three $ZZ(\gamma)$ gates per cost layer

Variant — the *alternating operator* ansatz.

Replace H_B with an XY mixer that preserves the number of 1s. One-hot constraints (L4) then hold by construction, with no penalty term.

E. Farhi, J. Goldstone, S. Gutmann (2014). "A quantum approximate optimization algorithm." arXiv:1411.4028 · S. Hadfield et al. (2019). "From the quantum approximate optimization algorithm to a quantum alternating operator ansatz." *Algorithms* 12, 34. arXiv:1709.03489

Common ansatz circuits (3): unitary coupled cluster

A physics-inspired ansatz from quantum chemistry. Start at the Hartree-Fock state and apply every single and double electron excitation, each with its own amplitude:

$$|\psi(\theta)\rangle = e^{T(\theta) - T(\theta)^\dagger} |\text{HF}\rangle$$

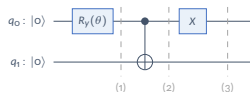
$$T(\theta) = \sum_{i,a} \theta_i^a a_a^\dagger a_i + \sum_{i<j, a<b} \theta_{ij}^{ab} a_a^\dagger a_b^\dagger a_j a_i$$

(i, j occupied orbitals; a, b empty ones). Keeping singles and doubles gives **UCCSD**.

- Conserves particle number by construction, so it never leaves the right sector
- $\theta = 0$ is Hartree-Fock, a physically sensible starting point
- Cost: $O(N^4)$ doubles for N spin orbitals, and the exponential must be Trotterized into long CNOT ladders. Deep on today's hardware

Smallest case: H_2 (L6, Example 3)

After tapering, only the double excitation survives and one parameter is left.



$$\begin{aligned} (1) \quad & \cos \frac{\theta}{2} |00\rangle + \sin \frac{\theta}{2} |10\rangle \\ (2) \quad & \cos \frac{\theta}{2} |00\rangle + \sin \frac{\theta}{2} |11\rangle \\ (3) \quad & \cos \frac{\theta}{2} |10\rangle + \sin \frac{\theta}{2} |01\rangle \end{aligned}$$

State (3) is exactly the L6 ansatz: $\theta = 0$ gives $|10\rangle$ (Hartree-Fock), and the optimum is $\theta^* = -14.4^\circ$.

A. Peruzzo et al. (2014). "A variational eigenvalue solver on a photonic quantum processor." *Nat. Commun.* 5, 4213 · J. Romero et al. (2018). "Strategies for quantum computing molecular energies using the unitary coupled cluster ansatz." *Quantum Sci. Technol.* 4, 014008.
arXiv:1701.02691

Common ansatz circuits compared

	Hardware-efficient	QAOA	UCCSD
Reference state	$ 0\rangle^{\otimes n}$	$ +\rangle^{\otimes n}$	$ \text{HF}\rangle$
Built from	native rotations and CNOTs	H_C (the problem) and H_B	fermionic excitations
Parameters m	$n(L+1)$ or $2n(L+1)$	$2p$	$O(N^4)$
Depth	shallow, chosen by you	set by the problem graph, $\times p$	deep after Trotterization
Symmetries kept	none	global bit flip, if H_C has it; Hamming weight with an XY mixer	particle number
Barren-plateau risk	high once deep	lower at small p	lower from Hartree–Fock
Used in	VQE on hardware, VQLS	combinatorial optimization (L7)	molecular VQE (L6)

- From left to right, more knowledge of the problem is built in. It buys fewer parameters and a better-conditioned landscape, and it costs depth
- Along the “Parameters” row, only QAOA’s count does not grow with the number of qubits

Takeaway: There is no best ansatz, only a best ansatz for a given problem, device and depth budget. Next: how the cost is measured, and how the parameters are optimized.

Measuring the cost

Hardware returns bitstrings measured in the Z basis.

Recipe

- 1 Write the observable as a sum of Pauli strings

$$H = \sum_{a=1}^L c_a P_a, \quad C(\theta) = \sum_a c_a \langle P_a \rangle_\theta$$

- 2 Rotate each non-Z factor before readout:
X: apply H Y: apply $S^\dagger H$
- 3 Estimate $\langle P_a \rangle$ from counts: each bitstring gives ± 1 (its parity)
- 4 **Group** strings that commute, so they share one circuit (a “setting”)

Shot noise

Error $\approx \sigma / \sqrt{M}$ with M shots. Accuracy ϵ on $C(\theta)$ needs

$$M \approx (\sum_a |c_a|)^2 / \epsilon^2.$$

Key points

- One evaluation = settings $\times$ shots per setting
- Halving ϵ costs $4 \times$ the shots
- Grouping cuts the number of circuits, not the $1/\epsilon^2$
- **Example:** H_2 (next frame), $\sum |c_a| = 1.47$,
 $\epsilon = 1.6 \times 10^{-3} \text{ Ha} \Rightarrow M \approx 8 \times 10^5$ shots per energy

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A. Peruzzo et al. (2014). *Nat. Commun.* 5, 4213 · D. Wecker, M. B. Hastings, M. Troyer (2015). “Progress towards practical quantum variational algorithms.” *Phys. Rev. A* 92, 042303 · V. Verteletskyi, T.-C. Yen, A. F. Izmaylov (2020). “Measurement optimization in the variational quantum eigensolver using a minimum clique cover.” *J. Chem. Phys.* 152, 124114

Measuring the cost: examples

Example 1 — Max-Cut on a triangle (one setting)

$H_C = \frac{1}{2} \sum_{(i,j)} (\mathbb{I} - Z_i Z_j)$ is diagonal: no rotations, measure once and average the cut value of each bitstring.

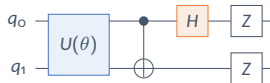
Bitstrings	Counts	Cut	Contribution
000, 111	50	0	0
the other six	950	2	1900
$\langle H_C \rangle = 1900/1000$			1.90 ± 0.014

QAOA also keeps the *best sample*: here any cut of value 2, which is optimal.

Example 2 — H_2 (two settings, L6)

$$H = q_0 + q_1 Z_0 + q_2 Z_1 + q_3 Z_0 Z_1 + q_4 Y_0 Y_1 + q_5 X_0 X_1$$

- **Setting A** (no extra gates):
 $\langle Z_0 \rangle$, $\langle Z_1 \rangle$, $\langle Z_0 Z_1 \rangle$ from the same counts
- **Setting B** (CNOT_{0→1}, then H on q_0):
 $\langle X_0 X_1 \rangle = \langle Z_0 \rangle_B$, $\langle Y_0 Y_1 \rangle = -\langle Z_0 Z_1 \rangle_B$



Check at Hartree-Fock $|10\rangle$: $\langle Z_0 \rangle = -1$, $\langle Z_1 \rangle = +1$,
 $\langle Z_0 Z_1 \rangle = -1$, $\langle XX \rangle = \langle YY \rangle = 0 \Rightarrow$
 $E = q_0 - q_1 + q_2 - q_3 = -1.1193$ Ha, as in L6.

—
 P. J. J. O'Malley et al. (2016). "Scalable quantum simulation of molecular energies." *Phys. Rev. X* 6, 031007 · A. Kandala et al. (2017). *Nature* 549, 242–246

Optimizing the parameters

The classical half of the loop. Given noisy estimates of $C(\theta)$, find $\theta^* = \arg \min_{\theta} C(\theta)$.

Two families

- **Gradient-free:** use only values of C (COBYLA, Nelder-Mead)
- **Gradient-based:** estimate ∇C on the device, then step

$$\theta \leftarrow \theta - \eta \nabla C(\theta)$$

Why not finite differences? Dividing shot noise by a small step h swamps the estimate.

K. Mitarai, M. Negoro, M. Kitagawa, K. Fujii (2018). "Quantum circuit learning." *Phys. Rev. A* 98, 032309 · M. Schuld, V. Bergholm, C. Gogolin, J. Izaac, N. Killoran (2019). "Evaluating analytic gradients on quantum hardware." *Phys. Rev. A* 99, 032331

Parameter-shift rule

For a gate $e^{-i\theta_k P/2}$ with $P^2 = \mathbb{I}$ (e.g. R_x, R_y, R_z):

$$\frac{\partial C}{\partial \theta_k} = \frac{1}{2} \left[C\left(\theta_k + \frac{\pi}{2}\right) - C\left(\theta_k - \frac{\pi}{2}\right) \right]$$

- **Exact**, and the large shift keeps it robust to noise
- Runs the *same* circuit with one angle changed
- **Cost:** $2m$ evaluations per gradient for m parameters

Common optimizers

Optimizer	Uses	Evaluations / step	Typical use
COBYLA, Nelder-Mead	values of C only	a few	simulators, few parameters
Gradient descent / Adam	parameter-shift gradient	$2m$	simulators, best final accuracy
SPSA	random 2-point estimate	2	noisy hardware (the default)
Quantum natural gradient	gradient + state geometry	$2m + \text{metric}$	fewer, costlier steps

SPSA in one line. Perturb *all* parameters at once along a random $\Delta \in \{\pm 1\}^m$:

$$\hat{g}_k = \frac{C(\theta + c\Delta) - C(\theta - c\Delta)}{2c \Delta_k}$$

Noisy, but correct on average, and only 2 evaluations whatever m is.

Rule of thumb

- Few parameters $\Rightarrow$ scan or gradient-free
- Many parameters, simulator $\Rightarrow$ Adam + parameter shift
- Many parameters, hardware $\Rightarrow$ SPSA

J. C. Spall (1992). *IEEE Trans. Automat. Control* 37, 332–341 · M. J. D. Powell (1994). COBYLA, in *Advances in Optimization and Numerical Analysis*, 51–67 · D. P. Kingma, J. Ba (2015). “Adam.” arXiv:1412.6980 · J. Stokes, J. Izaac, N. Killoran, G. Carleo (2020). “Quantum natural gradient.” *Quantum* 4, 269

Optimization: examples

Example 1 — parameter shift by hand

$|\psi(\theta)\rangle = R_Y(\theta) |0\rangle$, $C(\theta) = \langle Z \rangle = \cos \theta$. At $\theta = \pi/3$:

$$\frac{1}{2} \left[\underbrace{\cos \frac{5\pi}{6}}_{-0.866} - \underbrace{\cos(-\frac{\pi}{6})}_{0.866} \right] = -0.866 = -\sin \frac{\pi}{3} \checkmark$$

One gradient step with $\eta = 0.5$: $\theta : 1.047 \rightarrow 1.480$,
 $C : 0.50 \rightarrow 0.09$ (minimum: $C = -1$ at $\theta = \pi$).

Example 2 — H_2 from Hartree-Fock (L6)

At $\theta = 0$ the shift rule gives

$\partial E / \partial \theta = q_4 + q_5 = 0.185 > 0$, so the optimizer steps
to $\theta < 0$ and reaches $\theta^* = -14.4^\circ$, $E = -1.14268$ Ha.

Takeaway: Measurement sets the price of one evaluation; the optimizer sets how many you buy.

Example 3 — QAOA, $p = 1$ (L7)

Only $m = 2$ parameters (γ, β). Scan a coarse grid,
then refine the best point locally (e.g. COBYLA). A
 20×20 grid is 400 evaluations.

Example 4 — cost per iteration, $m = 100$

Hardware-efficient ansatz, 10^4 shots per evaluation:

	Evaluations	Shots
Parameter shift	200	2×10^6
SPSA	2	2×10^4

Over 200 iterations: 4×10^8 vs. 4×10^6 shots. This is why
SPSA is the hardware default.

Two objectives: expressibility and trainability

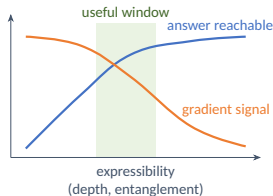
A good ansatz must meet **two objectives at once**.

1. Expressibility — can the ansatz reach the answer?

- How much of the state space the reachable set $\mathcal{M}$ covers
- Too low: the target is outside $\mathcal{M}$, so even a perfect optimizer returns a wrong answer (*ansatz error*)

2. Trainability — can the optimizer find it?

- Gradients large enough to measure with an affordable number of shots
- Too low: the landscape is flat, and the answer is in $\mathcal{M}$ but out of reach (*barren plateau*, next slide)



	Fails as	Check it by
Expressibility	ansatz error	compare with exact result, small n
Trainability	barren plateau	gradient variance vs. n

The conflict. The same knobs — depth, entanglement, lack of structure — raise expressibility and lower trainability.

Takeaway: Aim for an ansatz as expressive as the problem needs, and no more. Problem-inspired ansätze (QAOA, UCC) do this by being expressive only in the subspace where the answer lives.

S. Sim, P. D. Johnson, A. Aspuru-Guzik (2019). "Expressibility and entangling capability of parameterized quantum circuits for hybrid quantum-classical algorithms." *Adv. Quantum Technol.* 2, 1900070 · Z. Holmes, K. Sharma, M. Cerezo, P. J. Coles (2022). *PRX Quantum* 3, 010313, R. M. Sanz, A. Anglès-Castillo, E. Alarcón and C. G. Almudéver, "Efficiently Architecting VQAs: Expressibility-Trainability-Resources Pareto-Optimality," in *IEEE Transactions on Quantum Engineering*, vol. 7, pp. 3104517-3104517, 2026, Art no. 3104517, doi: 10.1109/TQE.2026.3721256.

Barren plateaus

What. For random θ , gradients vanish exponentially in the qubit count n : $\text{Var}[\partial C/\partial \theta_k] \sim b^{-n}$, $b > 1$. Seeing a slope needs $\sim b^n$ shots ($n = 50$: $\text{Var} \approx 10^{-15}$).

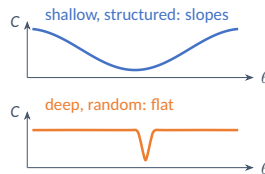
Why. *Expressibility* = how uniformly the ansatz covers all states. A highly expressive ansatz acts like a random unitary, so $C(\theta)$ is flat almost everywhere.

Expressibility factors

- **Depth:** more layers, closer to random
- **No structure:** generic hardware-efficient gates
- **Random start:** begins in the flat region
- **Heavy entanglement:** each qubit alone looks random

Other causes: **global** costs and **hardware noise**.

Takeaway: A barren plateau is not slow convergence — there is no gradient to follow.



Cause	Mitigation
Depth, no structure	shallow, problem-inspired ansatz
Random start	start near Hartree-Fock or identity
Global cost	local cost terms

J. R. McClean et al. (2018). *Nat. Commun.* 9, 4812 · Z. Holmes, K. Sharma, M. Cerezo, P. J. Coles (2022). *PRX Quantum* 3, 010313 · M. Cerezo et al. (2021). *Nat. Commun.* 12, 1791 · S. Wang et al. (2021). *Nat. Commun.* 12, 6961

Noise, and how it changes the problem

Every claim above assumed exact expectation values. On hardware:

- **Shot noise** puts a floor under the cost. With L measured terms and M shots the floor is $\mathcal{O}(L/\sqrt{M})$ — halving it costs four times the shots.
- **Gate noise** biases the cost, so the optimizer converges to the wrong point, not merely a noisy one.
- **Noise-induced plateaus** flatten the landscape independently of the ansatz; beyond a depth threshold the state approaches the maximally mixed state.

Error mitigation buys accuracy with shots, not depth.

Technique	What it does	Cost
Readout calibration	inverts the measurement confusion matrix	cheap, always worth it
Zero-noise extrapolation	runs at amplified noise, extrapolates to zero	$3-5 \times$ shots
Probabilistic error cancellation	unbiased estimator by quasi-probability sampling	exponential in circuit volume

Before you run a VQA*

Five diagnostics, in this order. Four of them are classical computations you can do on a laptop before asking for device time.

Diagnostic	How	Stop if
Gradient variance	sample $\sim 10^2$ random θ , take one parameter-shift derivative, repeat at $n = 4, 6, 8, 10$ and fit $\log \text{Var}$ against n	the slope is steeply negative — the ansatz has a plateau and no optimizer will rescue it
Classical simulability	run the same ansatz as a matrix-product state at modest bond dimension, or with a Clifford + perturbation method	it converges: your quantum run would demonstrate nothing
Shot budget	$(\sum_a c_a)^2 / \epsilon^2 \times 2m \times \text{iterations}$	it exceeds your allocation — redesign the cost or the ansatz, not the shot count
Classical baseline	run the best classical heuristic on the identical instance	you cannot state the number you have to beat
Noise floor	run the ansatz at parameters whose exact answer you already know	the mitigated value is still off by more than your target ϵ

Takeaway: Running these afterwards is how a result gets published and then reproduced on a laptop. Running them first costs a day and tells you whether the experiment can succeed.

Next

- **L6** — the first workhorse instance: VQE for ground states, worked end to end, then the variational quantum linear solver for $A\mathbf{x} = \mathbf{b}$
- **L7** — the second: QAOA, the instance built specifically for combinatorial optimization
- **L8-L9** — putting the linear solver to work on partial differential equations

References

Grover adaptive search

- C. Dürr & P. Høyer (1996). "A Quantum Algorithm for Finding the Minimum." *arXiv:quant-ph/9607014* (preprint)
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The variational framework

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